# GWAS analysis of human height

**Genomic Thinking Project**

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## Abstract

This study summarizes a genome-wide association analysis (GWAS) on a diverse dataset focusing on the phenotype of height using publicly available data from OpenSNP database. Specifically, the dataset came from different users and companies, therefore it went through a quality control (QC) and population structure assessment using PLINK 1.9 and PCA [1], [2]. Afterwards, GWAS was performed using both simple and mixed linear models. For the mixed models, the GCTA software tool was used [3]. Despite several optimizations, no significant SNPs were identified in connection with human height. Lastly, half of the data was used as a training set to calculate a polygenic score (PGS) for the height and then check whether this PGS shows any significance on the validation set [4]. It turned out that the PGS doesn't account for a significant portion of height variation. These results highlight the challenges of working with a diverse dataset while doing GWAS analysis.

## Introduction

The OpenSNP website aggregates whole genome sequences from different companies and individuals. There are advantages and disadvantages to running a GWAS analysis on such a dataset. On one hand, distant and unrelated individuals would mean that there is no population structure for the dataset. On the other hand, the dataset may be diverse, since there are different companies using different sequencer devices, therefore one SNP might be present for company X but not company Y. Regarding the height, which is also a complicated phenotype, previous scientific work has shown that there are many SNPs, which might be related to how height varies amongst different populations [4]. Besides the genetic variation, environment also plays a significant role in how tall an individual is. Keeping these factors in mind, I developed a customizable workflow with steps for analyzing a diverse dataset using Snakemake workflow manager [5]. This study aims to find relevant genes and SNPs that correlate with the phenotype of height and investigate how PGS can be utilized as a predictor for human height.

## Discussion

The workflow consists of several stages and steps, which are summarized in Figure 1. At first, a sample-based QC was applied to the dataset, which filtered out individuals but not variants. This stage started with a sex check, it should be noted that some entries in the FAM file didn't contain sex information, thus the sex check was run with the "--allow-no-sex" flag. Although we didn't have sex information for some of the entries, PLINK could infer it based on X chromosome homozygosity [6]. Afterward, a missingness rate per individual was applied, but almost all the entries had a missingness proportion of above than 0.5, thus there was no filtering applied. In the next step, heterozygosity was investigated and filtered out the entries that had 3 standard deviations for both sides from the mean. Lastly, identity by descent (IBD) was checked to identify and remove related individuals, which could undermine the GWAS analysis. Before calculating the IBD, the data was pruned by getting rid of SNP pairs that had $R^2$ value of greater than or equal to 0.2 (typically chosen threshold), using 500kb as window size and 5 variants as step size. IBD was then calculated using the "--min 0.185" option, and eventually removed these entries from the data. Note that third-degree relatives have an IBD of 0.125. The above steps included the sample-based QC, the number of filtered-out individuals is shown in Figure 2 on the left.

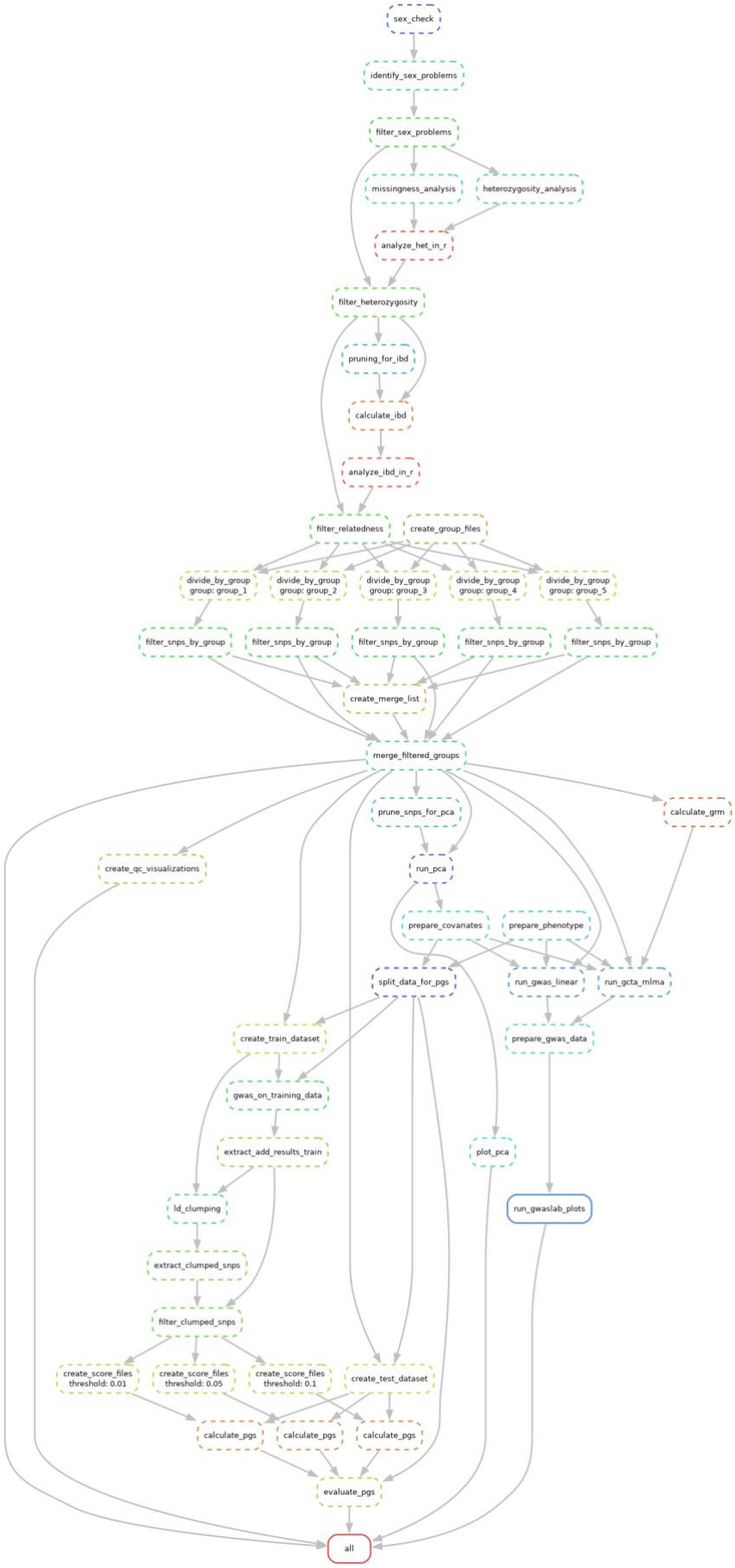


*Figure 1 Workflow steps made by Snakemake [5].*

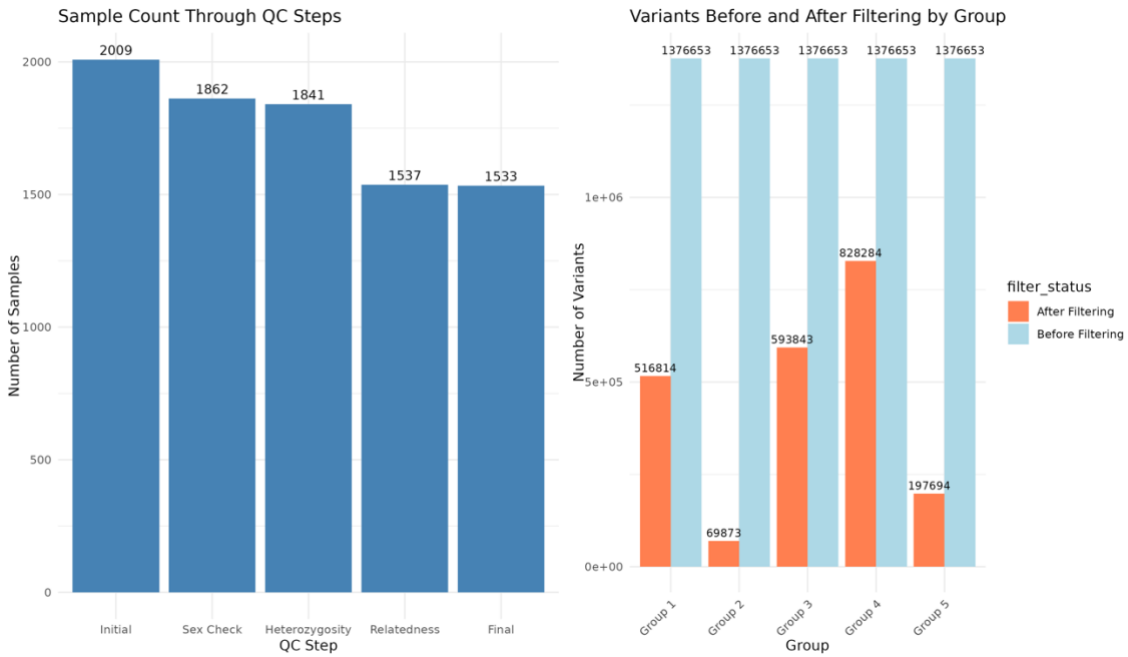


*Figure 2 Visualization of QC. On the left sample-based QC can be seen, while on the right SNP QC is shown.*

The next stage was a SNP quality control, where variants were removed rather than samples. Because of the diversity of the dataset, the SNP QC was only applied after the data had been separated based on the metadata, specifically the chips. In total 5 groups were identified out of which 4 had several samples above than 50, Group 2 included only 4 individuals, thus it was removed in order not to make the merging step worse onward. Missingness rate, Hardy-Weinberg Equilibrium, and minor allele frequency filters were applied with the standard threshold values, the result can be seen in Figure 2 on the right [1]. The QC workflow stage ended by merging the 4 remaining groups. After the QC was finished the BIM file contained 896,978 variants, while there were 1,533 samples in the FAM file. In the next step, a PCA was performed to check whether there is a population stratification present in the QC data. At first, I made an LD-pruned set of SNPs and then used it to calculate the first 20 principal components (PCs). Both the scree plot and the PCA plots (Figure 3) indicate significant population structure within the samples.

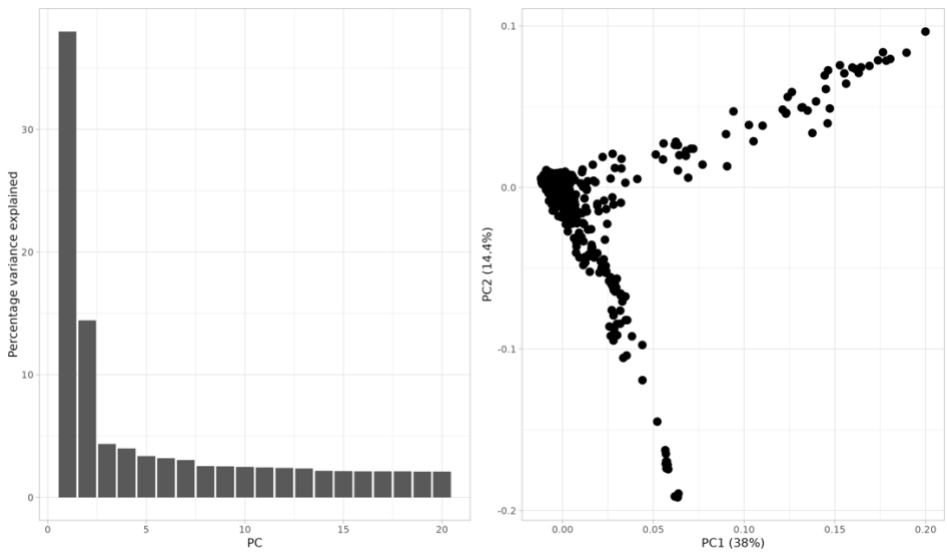


*Figure 3 Scree (left) and PCA (right) plots.*

The first two principal components mostly capture the genetic variation, likely reflecting the diversity of the dataset. My intuition from the PCA plot is that it verifies the fact that the dataset comes from two different companies. Since there is a population structure present in the QC data, the onward analyses used the covariates (first 10 PCs) to control for the confounding. In the next stage phenotype and covariate files were built to run the GWAS analysis. I used two different models for the GWAS, a simple linear model from PLINK [2] and a mixed linear model from GCTA [3]. The results can be seen in Figure 4 and Figure 5. Using a simple linear model, the result is not significant for any of the SNPs, although the QQ plot seems fine, with genomic inflation factor being close to 1 means that the calibration of p-values is well done. The tail of the QQ plot doesn't deviate that much, there are some points, which indicate significance, but altogether none of the SNPs reached conventional GWAS significance ($p < 5 \times 10^{-8}$) as it can be seen in Figure 4.

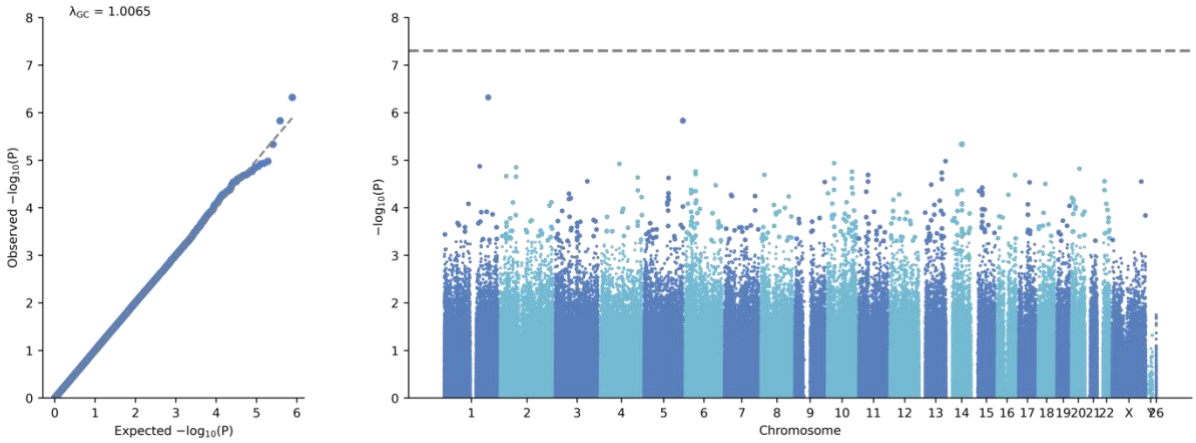


*Figure 4 Result of GWAS using simple linear model [7].*

In Figure 5 the result of the GWAS using mixed linear model can be seen, on the left the QQ plot shows that most of the SNPs follow the expected p-values, thus we can say that it is well controlled, there are some tail deviations, indicating a small number of SNPs being significant. There is one SNP that reached the threshold value mentioned previously, it can be seen on the right Manhattan plot below.

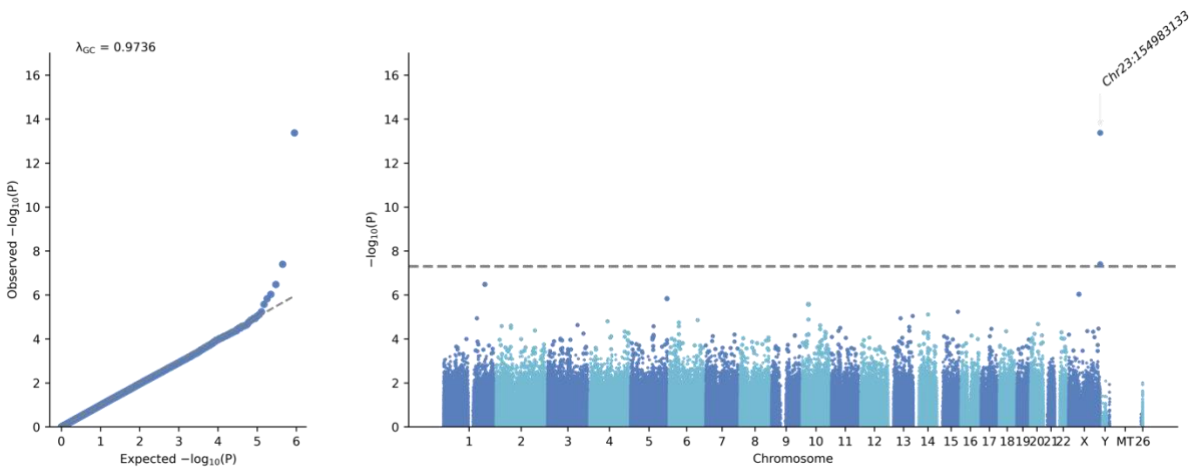


*Figure 5 Result of GWAS using mixed linear model [7].*

The analysis ended with a polygenic score (PGS) prediction, by splitting up the QC data into a training set (70%) and a validation set (30%). Afterward, a GWAS was calculated based on the training set, using simple linear model. This was followed by an LD-clumping process where independent SNPs and LD structure were identified, which was then used to filter out the GWAS result creating score files for different thresholds (0.01, 0.05, 0.1). These score files were used to create PGS on the validation set. The threshold of $p < 0.01$ demonstrated the highest predictive accuracy among the above thresholds, based on variance explained in the testing set. To assess the statistical significance, I fitted a linear model to the PGS and the 10 PCs against the height, the association between the height and the PGS (Figure 6) turned out to be insignificant ($p = 0.616$). While the PGS calculated using the $P < 0.01$ threshold explained the most variance among the tested thresholds, this predictive ability was not statistically significant in the independent test set when principal components were present.

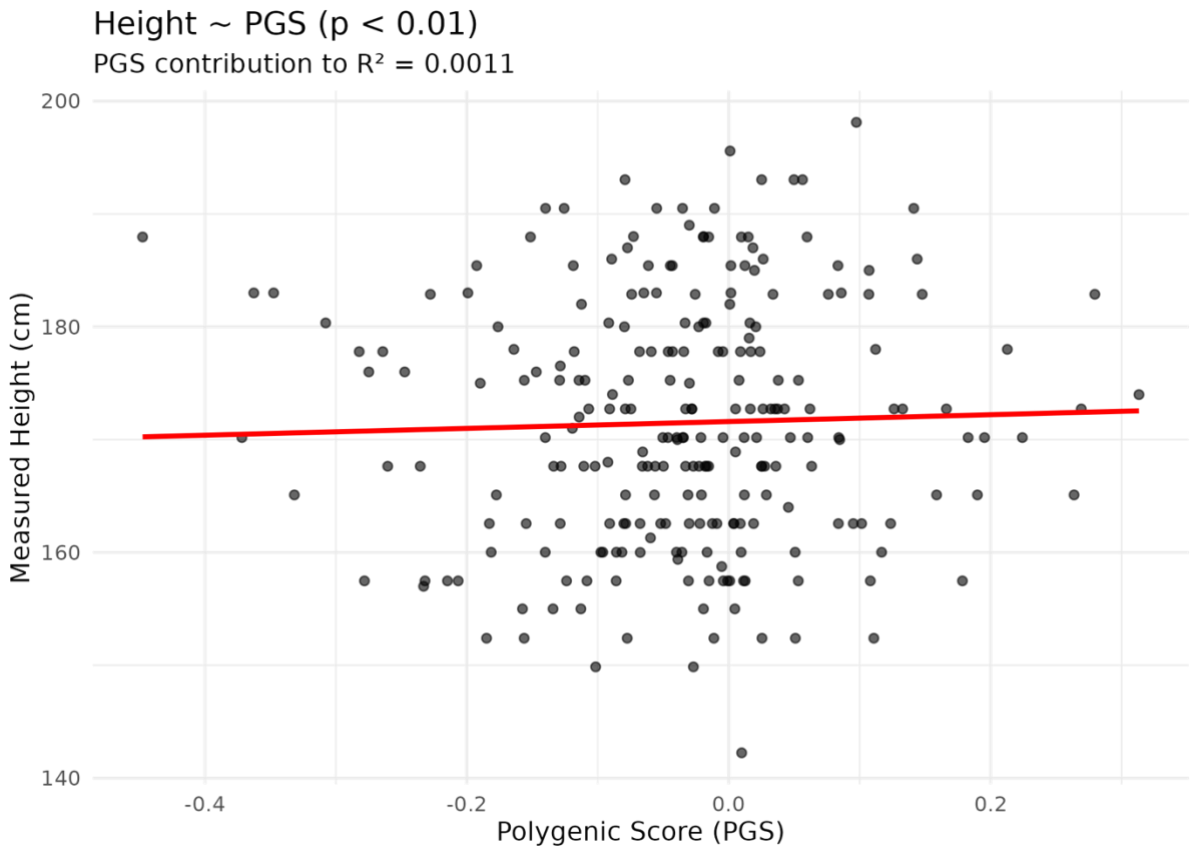


*Figure 6 Result of PGS prediction.*

## Conclusion

Overall, this study showed the challanges of performing GWAS and PGS analysis on a diverse dataset from OpenSNP. Despite strict quality control and using both simple and mixed linear models, no SNPs were found to have significant association with height. The PGS prediction also failed to significantly predict height variation in the validation set. These findings highlight the problems of working with heterogeneous genetic data. All the code can be found in the linked reference [8].

## References


[1] C. A. Anderson, F. H. Pettersson, G. M. Clarke, L. R. Cardon, A. P. Morris, and K. T. Zondervan, “Data quality control in genetic case-control association studies,” *Nat Protoc*, vol. 5, pp. 1564–1573, Aug. 2010, doi: 10.1038/nprot.2010.116.

[2] C. C. Chang, C. C. Chow, L. C. A. M. Tellier, S. Vattikuti, S. M. Purcell, and J. J. Lee, “Second-generation PLINK: Rising to the challenge of larger and richer datasets,” *Gigascience*, vol. 4, Feb. 2015, doi: 10.1186/s13742-015-0047-8.

[3] J. Yang, S. H. Lee, M. E. Goddard, and P. M. Visscher, “GCTA: A tool for genome-wide complex trait analysis,” *Am J Hum Genet*, vol. 88, pp. 76–82, Jan. 2011, doi: 10.1016/j.ajhg.2010.11.011.

[4] L. Yengo *et al.*, “A saturated map of common genetic variants associated with human height,” *Nature*, vol. 610, pp. 704–712, Oct. 2022, doi: 10.1038/s41586-022-05275-y.

[5] F. Mölder *et al.*, “Sustainable data analysis with Snakemake,” *F1000Res*, vol. 10, p. 33, Jan. 2021, doi: 10.12688/f1000research.29032.1.

[6] “PLINK 1.9 beta,” https://www.cog-genomics.org/plink2/.

[7] Y. He, M. Koido, Y. Shimmori, and Y. Kamatani, “GWASLab: a Python package for processing and visualizing GWAS summary statisticsの表示,” https://jxiv.jst.go.jp/index.php/jxiv/preprint/view/370/1223.

[8] “markoc1120/gwas_analysis: Genomic Thinking final project,” https://github.com/markoc1120/gwas_analysis.